

Real-time Airfare Price Index (APIx)

Comprehensive System Architecture & Implementation Strategy: Augmenting the Consumer Price Index (CPI) through Automated, High-Frequency Web Scraping of Airline & Aggregator Platforms.

Core Objective: Replace obsolete manual ticket counter surveying with an automated, high-frequency data ingestion engine that mirrors actual consumer booking realities across India's domestic aviation network to provide RBI and NSO with a precise, real-time inflation metric.

1. The Problem Space & Empirical Findings

Currently, the National Statistical Office (NSO) calculates the "Transport and Communication" sub-index using manual price quotes gathered from physical airline ticketing counters. However, over **92% of domestic Indian air tickets** are purchased via Online Travel Aggregators (OTAs) and carrier web portals. This structural divergence introduces systemic measurement errors into the Consumer Price Index (CPI).



The "Drip Pricing" Disturbance

Analysis of Indian travel forums (e.g., *r/AirTravelIndia*, *r/CreditCardsIndia*) reveals that baseline search prices frequently exclude mandatory surcharges: Convenience Fees (₹350–₹550), CUTE/RCS fees, and seat assignment levies. Current price tracking captures unrepresentative base figures, under-reporting out-of-pocket ticket inflation by **8% to 15%**.



High-Frequency Dynamic Spikes

Algorithmic yield management software dynamically updates ticket prices based on real-time booking velocity and inventory thresholds. Single-day fare variations routinely exceed **200%–400%** between off-peak mid-day windows and evening peak slots on primary trunk sectors like DEL–BOM and BLR–DEL.



Phantom Inventory & Caching Lag

Aggregator portals rely on cached Global Distribution System (GDS) responses to optimize load times. When consumers proceed to payment, up to **14% of low-cost quotes fail**, triggering immediate price re-evaluations. Scraping front-end search cards captures non-existent, stale rates.



Promotional Distortions & Bundling

Aggregators artificially inflate nominal convenience fees while offering targeted bank discounts (e.g., ICICI/HDFC instant codes). An authoritative index must decouple net consumer outlays from synthetic promotional credit structures.

2. Target Stakeholder Utility

Beneficiary	Operational Function	Direct Strategic Impact
NSO (MoSPI)	CPI Sub-Index Production	Transitions national accounts from lag-heavy surveys to an automated, auditable daily time-series.
Reserve Bank of India	Monetary Policy Formulation	Captures real-time demand-pull and energy cost-push inflationary pressures with zero reporting lag.
DGCA / Min. of Civil Aviation	Sector Oversight & Tariff Tracking	Monitors anti-competitive fare surges during operational disruptions, festival seasons, and extreme weather.

3. Comparative Matrix: Existing Solutions vs. APIx

Existing flight tracking platforms prioritize consumer conversions or commercial price intelligence rather than rigorous econometric standards.

System / Solution	Target Problem	Underlying Flaw / Loophole	Why the Gap Exists	APIx Architectural Countermeasure
Google Flights / Skyscanner	Meta-search and lead generation for booking portals.	Omits mandatory checkout convenience charges and payment-gateway fees.	Architected for sub-second query latency, bypassing final checkout stage steps.	Multi-stage scraping pipeline tracks full checkout path, isolating base fare from mandatory surcharges.
Commercial Scrapers (Apify, Bright Data)	Ad-hoc commercial price intelligence for travel portals.	Scrapes generic top-level search cards; fails on dynamic seat bundles and fare families.	High compute costs prevent full checkout traversal across deep advance booking windows.	Direct network microservice API interrogation extracting structured fare tiers (Saver, Flexi, Corporate).
Hopper / Ixigo Price Predictor	Consumer purchase timing and price freeze upsells.	Black-box proprietary algorithms disconnected from international index methodology.	Designed to maximize ancillary margin and fintech subscription revenue.	Fully transparent, open-methodology chained Laspeyres/Tornqvist index formulas compliant with IMF standards.
Traditional MoSPI Manual Survey	Official retail inflation measurement.	Fixed monthly physical counter sampling; misses 90%+ digital transactions and surge pricing.	Legacy operational reliance on offline field investigator frameworks.	Scheduled, automated cloud scrapers capturing fixed advance purchase horizons (T+1 to T+45) daily.

4. Operating Model & System Archetype Comparison

Commercial Booking Platforms (OTAs) Objective: Gross Booking Volume (GBV) and commission maximization. Revenue Mechanism: 2%–6% airline commissions, non-refundable platform fees, and paid placements. Data Philosophy: Cached, approximate pricing calibrated to push searchers into the conversion funnel.	APIx Sovereign Infrastructure (NSO / RBI) Objective: Statistical accuracy, econometric rigor, and auditability. Funding Model: Public digital infrastructure grant / MoSPI statistical budget. Data Philosophy: Unbiased, immutable time-series data with detailed breakdown of base tariffs and government levies.
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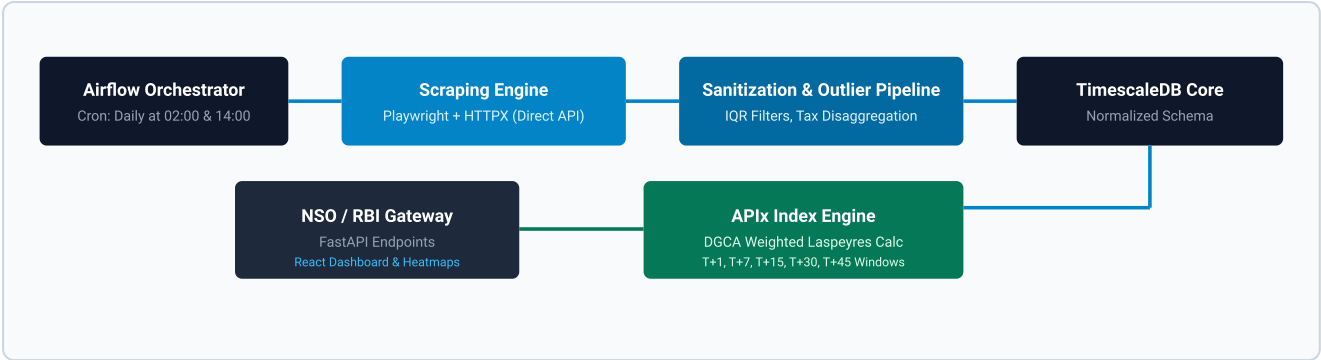
5. Target Data Sourcing Hierarchy

To ensure statistical validity, data collection follows a structured, priority-weighted hierarchy:

- **Tier-1: Carrier-Direct Portals (75% Weight):** InterGlobe Aviation (IndiGo - 62% market share), Air India Group (Air India & AI Express), Akasa Air, and SpiceJet. Direct mobile endpoint scraping extracts pure fare families (Saver vs. Flexi), fuel surcharges (YQ), and statutory fees (UDF/PSF/GST).
- **Tier-2: Leading Aggregators (25% Weight):** MakeMyTrip, EaseMyTrip, and Ixigo. Included specifically to monitor platform-level mandatory convenience fees and capture multi-carrier interline pricing.

6. End-to-End Architectural Pipeline

The APIx architecture uses a modular microservices model built for reliable data collection, validation, and index generation.



7. Scraping Resilience & Ethical Governance

 **Anti-Bot & Session Management**

- **Direct API Interception:** Bypasses heavy DOM rendering by sniffing mobile endpoint Bearer tokens and querying backend flight-search JSON endpoints directly.
- **TLS Fingerprint Randomization:** Emulates standard browser JA3/JA4 TLS signatures using `curl_cffi` to prevent Cloudflare/Akamai bot detection.
- **Residential IP Rotation:** Distributes requests across regional Indian IP pools with automatic rate-limit backoffs.

 **Ethical Scraping Safeguards**

- **Robots.txt & SLA Compliance:** Extraction routines run during off-peak hours (01:00 to 04:00 IST) to minimize host server utilization.
- **Strict Rate-Limitation:** Enforces minimum 3.5s jittered delays between sector queries; zero concurrent hits on single airline domain endpoints.
- **Identifiable User-Agents:** Outlines administrative research contact details in custom request headers for platform visibility.

8. Data Cleaning & Disaggregation Model

Every collected quote is processed through a multi-tier sanitization pipeline:

- **Outlier Elimination:** Applies a modified Interquartile Range (IQR) filter ($\$1.5 \times \text{IQR}$) per route-advance window to drop erroneous baseline data while preserving legitimate holiday demand surges.
- **Fare Unbundling:** Programmatically separates total outlay into: **Total Fare = Base Fare + Fuel Surcharge (YQ) + Airport Fees (UDF/PSF) + GST + Mandatory Platform Fee.**
- **Phantom Ticket Filter:** Flags and discards quotes where inventory states indicate < 2 seats remaining at non-reproducible promotional rates.

9. Index Construction Methodology

The Airfare Price Index (APIx) utilizes a **Chained DGCA-Weighted Laspeyres Index**, integrating route-level passenger traffic shares and multi-horizon advance purchase allocations.

Mathematical Definition:

$$APIx_t = 100 \times \sum_{r \in R} \left[W_r \times \sum_{k \in K} \left(\omega_k \times \frac{P_{\{r,k,t\}}}{P_{\{r,k,0\}}} \right) \right]$$

Where W_r is the official DGCA passenger share of route r , ω_k is the empirical weight of advance-purchase window $k \in \{T+1, T+7, T+15, T+30, T+45\}$, and $P_{\{r,k,t\}}$ is the geometric mean fare for that cell on day t .

10. Representative Route Basket (DGCA Top Sectors)

City Pair	Sector Type	Monthly Pax Share (%)	Advance Windows Monitored	Carriers Sampled
DEL – BOM	Primary Metro Trunk	12.4%	T+1, T+7, T+15, T+30, T+45	IndiGo, Air India, Akasa, SpiceJet
DEL – BLR	Tech Corridor / Trunk	9.8%	T+1, T+7, T+15, T+30, T+45	IndiGo, Air India, Akasa
BOM – BLR	Inter-Metro High Density	7.6%	T+1, T+7, T+15, T+30	IndiGo, Air India, Akasa
DEL – CCU	East Metro Trunk	5.2%	T+1, T+7, T+15, T+30, T+45	IndiGo, Air India, SpiceJet
BLR – HYD	Regional Short-Haul	4.3%	T+1, T+7, T+15, T+30	IndiGo, Air India Express

11. 30-Day Back-Testing & Validation Protocol

To ensure macroeconomic reliability, the APIx engine will back-test its aggregated monthly series against historical DGCA average-fare data. The target acceptance threshold is a Pearson correlation coefficient of $R^2 \geq 0.88$ across primary metro sectors.

12. Prototype Deliverables & Core Feature Checklist

Core Backend & Data Pipeline - Scraping engine targeting IndiGo, Air India, Akasa, and MakeMyTrip. - Automated scheduled ingestion for 5 standard DGCA routes. - Outlier removal and automated fare unbundling pipeline. - PostgreSQL / TimescaleDB storage with full audit logs.	Analytical UI & Export API - Daily, weekly, and monthly APIx interactive time-series plots. - Route-level price heatmaps and advance purchase elasticity curves. - Role-based access dashboard for MoSPI / RBI analysts. - REST API delivering aggregated inflation time-series in JSON/CSV.
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